REPUBLIC POLYTECHNIC

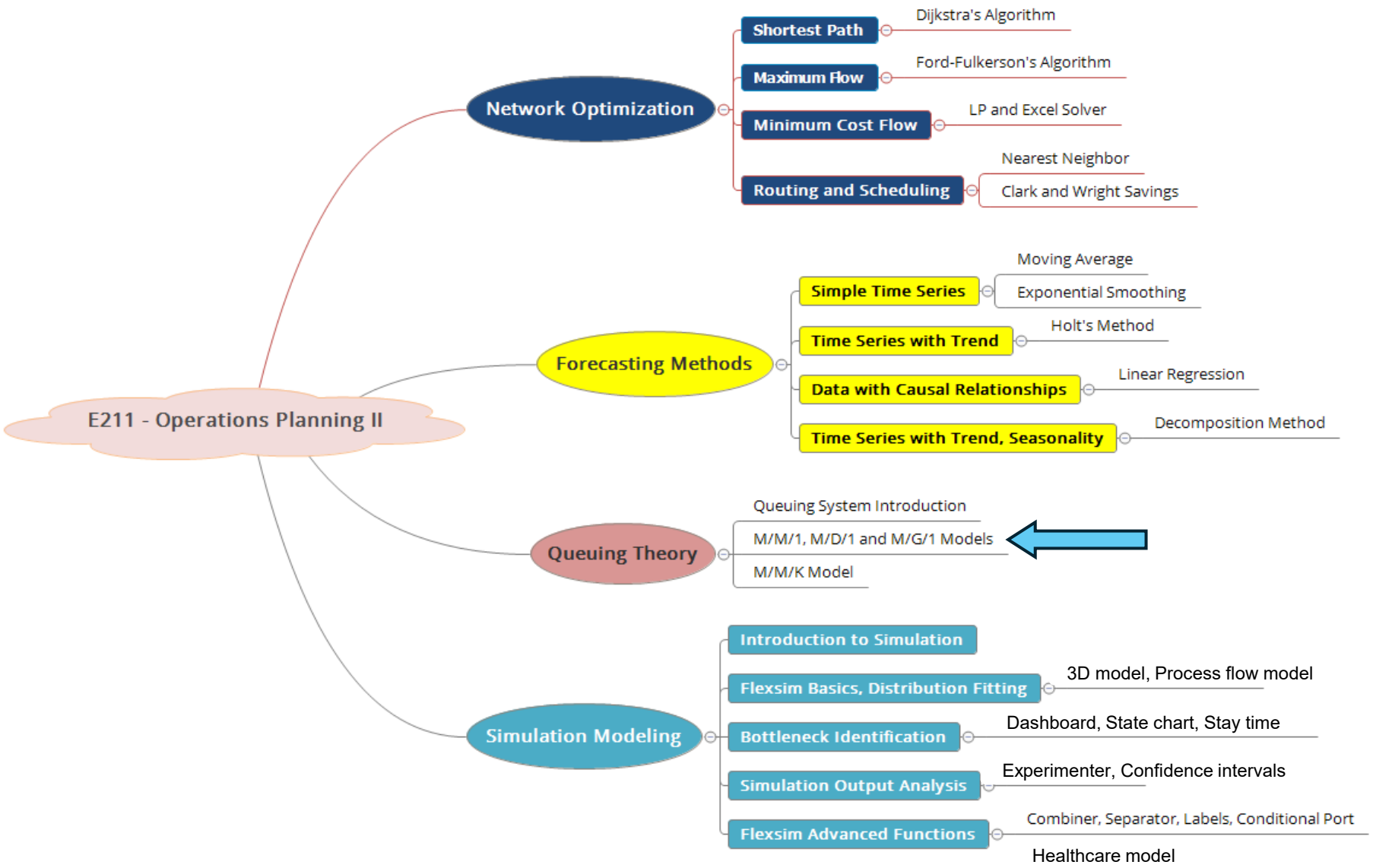
Lesson 2

Queuing Theory – Single Server Models (M/M/1, M/G/1, M/D/1)

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

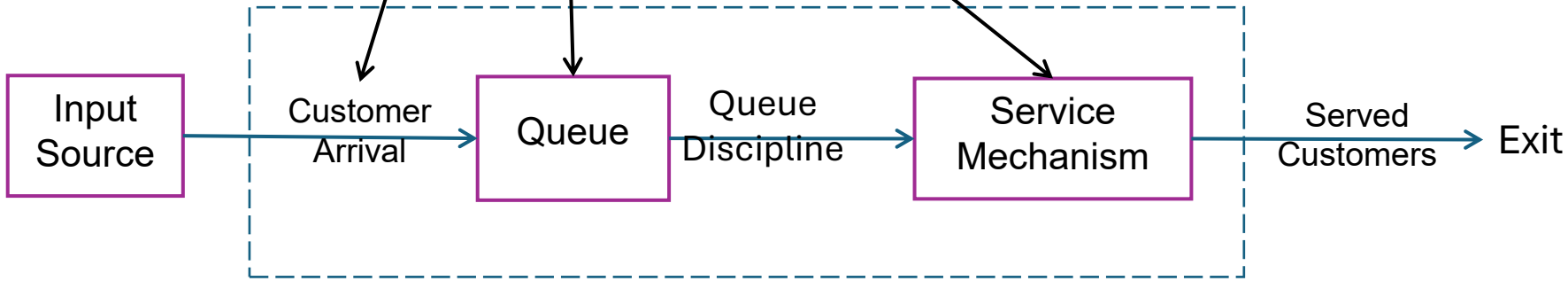
E211 Topic Tree



Queuing Models

Queuing Process

Queuing system can be described with **Arrival**, **Queue** and **Service** components



How well is the queuing system performing?

Purpose of Queuing Models

Queuing models are mathematical models used to:

- Describe the **behaviour** of queuing systems

- Determine the **performance** of the queuing systems. Example:
  - ✓ Average number of customers in system
  - ✓ Average waiting time in queue

- Determine the **service level** to provision. Example:
  - ✓ Number of servers to provide
  - ✓ Average service time for each customer

Kendall's Notation for Queuing Models

- In general, **Kendall** classification of queuing systems classification uses 6 symbols (parameters):

A/B/C/D/E/F

- **A** – Arrival pattern (distribution of intervals between arrivals).
  - M: Poisson (Markovian) process with exponential distribution of intervals
  - D: Deterministic(Constant) inter-arrival times
  - G: Inter-arrival time follows a general (any) distribution
- **B** – Service pattern (distribution of service duration).
  - M: Poisson (Markovian) process with exponential distribution of service duration.
  - G: Service duration follows a general (any) distribution
  - D: Deterministic(Constant) service duration
- **C** – Number of servers

Kendall's Notation for Queuing Models

A/B/C/D/E/F

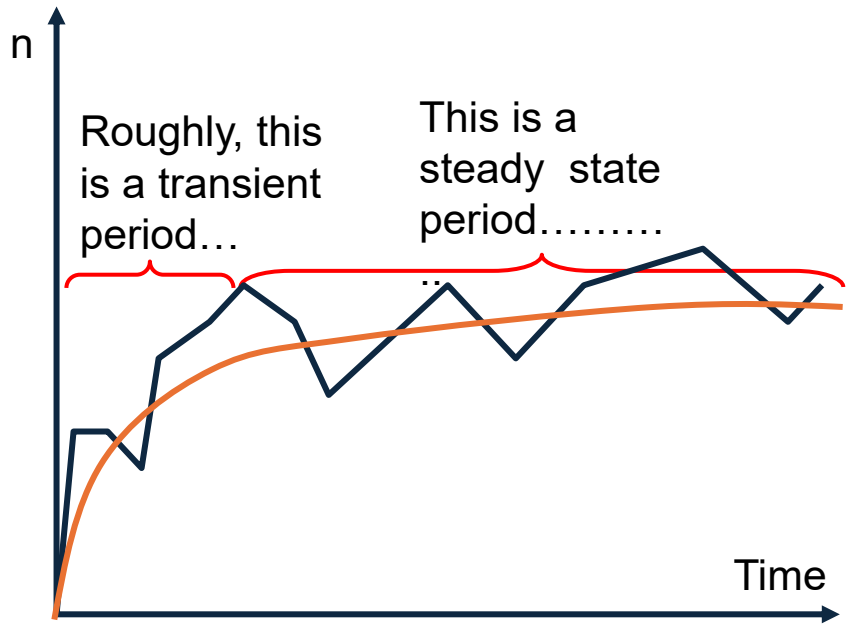
- **D** – Queue discipline (FIFO, LIFO, ...). Omitted for FIFO or if not specified.
- **E** – System capacity. Omitted for unlimited queue length.
- **F** – Population size (number of possible customers). Omitted for infinite systems.

Examples

- **D/M/1/LIFO/10/50**: Deterministic (known) arrivals, one server with exponentially distributed service time, queue is limited by a maximum size of 10 with a last-in-first-out queueing discipline, and the total number of customers is 50.
- **M/G/3**: Multiple-server (3 servers) queuing model with Poisson (random) arrivals, service time following a general distribution, FIFO queuing discipline, infinite number of waiting positions, and unlimited customer population.

Performance Measures (I)

Initial (transient) behaviour may not be representative of long term performance



Queue Performance is measured at <u>steady state</u>.

EQUILIBRIUM CONDITION:

To reach steady state, effective arrival rate (λ) must be less than sum of effective service rates (μ).

$\lambda < \mu$ (for single server)

$\lambda < K\mu$ (for K servers)

Performance Measures (II)

These are some common performance measures that can be obtained from queuing models:

- System utilization, ρ **(Rho)**
- Probability of idle service facility, P_0
- Probability of *n* entities in system, P_n
- Average queue time, **W*q***
- Average queue length, **L*q***
- Average time in system, **W*s***
- Average number in system, **L*s***

M/M/1 Queuing Model

M/M/1 Model Characteristics

- Input source: Infinite; no balks, no reneging
- Arrival process: Number of arrivals follows **Poisson distribution**
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service mechanism: **Single server**, single phase system
- Service time: **Negative Exponential** distribution
- Arrivals and service times are independent of the number of customers in system
- Service rate > arrival rate

Poisson Distribution

- Poisson distribution describes the **number of events (arrivals) that occur randomly in an interval of time**
- Negative exponential distribution describes the **service time distribution in terms of time between (service) events**
- If customer arrivals follow Poisson distribution, then time between such arrivals (inter-arrival time) follows an exponential distribution.
- If service time follows exponential distribution, then number of customers served follows Poisson distribution.
- Both Poisson and negative exponential distributions are commonly used to study queuing processes that exhibit random arrival and service times property

General Equations for Single Server Models

- Equations applicable to <u>single server</u> queuing models with **M**arkovian (Poisson) arrival process, e.g. M/M/1, M/G/1 and M/D/1.

Arrival rate

System Utilization (Rho), $$\rho = \frac{\lambda}{\mu}$$

Service rate

Probability of no entity in system, $$P_o = (1-\rho)$$

Probability of n entities in system, $$P_n = \rho^n (1-\rho)$$

M/M/1 Equations

Average number of entities waiting for service,

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{\rho^2}{1-\rho}$$

Little's Formula

Average time an entity spends in the queue,

$$W_q = \frac{L_q}{\lambda}$$

Average number of entities in the system,

$$\mathrm{L_s} = \mathrm{L_q} + \frac{\lambda}{\mu}$$

Average time an entity spends in the system,

$$\mathrm{W_s} = \mathrm{W_q} + \frac{1}{\mu}$$

M/G/1 Queuing Model

M/G/1 Model Characteristics

- Input source: Infinite; no balks, no reneging
- Arrival process: Number of arrivals follows **Poisson distribution**
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service mechanism: **Single server**, single phase system
- Service time: Any **General Distribution** with mean μ and standard deviation σ
- Arrivals and service times are independent of the number of customers in system
- Service rate > arrival rate

M/G/1 Equations

Standard deviation of service times

Average number of entities waiting for service,

$$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)}$$

Little's Formula

Average time an entity spends in the queue,

$$W_q = L_q / \lambda$$

Average number of entities in the system,

$$L_s = L_q + \frac{\lambda}{\mu}$$

Average time an entity spends in the system,

$$W_s = W_q + \frac{1}{\mu}$$

M/D/1 Queuing Model

M/D/1 Model Characteristics

- Input source: Infinite; no balks, no reneging
- Arrival process: Number of arrivals follows **Poisson distribution**
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service mechanism: **Single server**, single phase system
- Service time: **Constant (deterministic)**
- Arrivals and service times are independent of the number of customers in system
- Service rate > arrival rate

M/D/1 Equations

Average number of entities waiting for service,

$$L_q = \frac{\rho^2}{2(1-\rho)}$$

Little's Formula

Average time an entity spends in the queue,

$$W_q = L_q / \lambda$$

Average number of entities in the system,

$$L_s = L_q + \frac{\lambda}{\mu}$$

Average time an entity spends in the system,

$$W_s = W_q + \frac{1}{\mu}$$

Comparison of Single Server Models

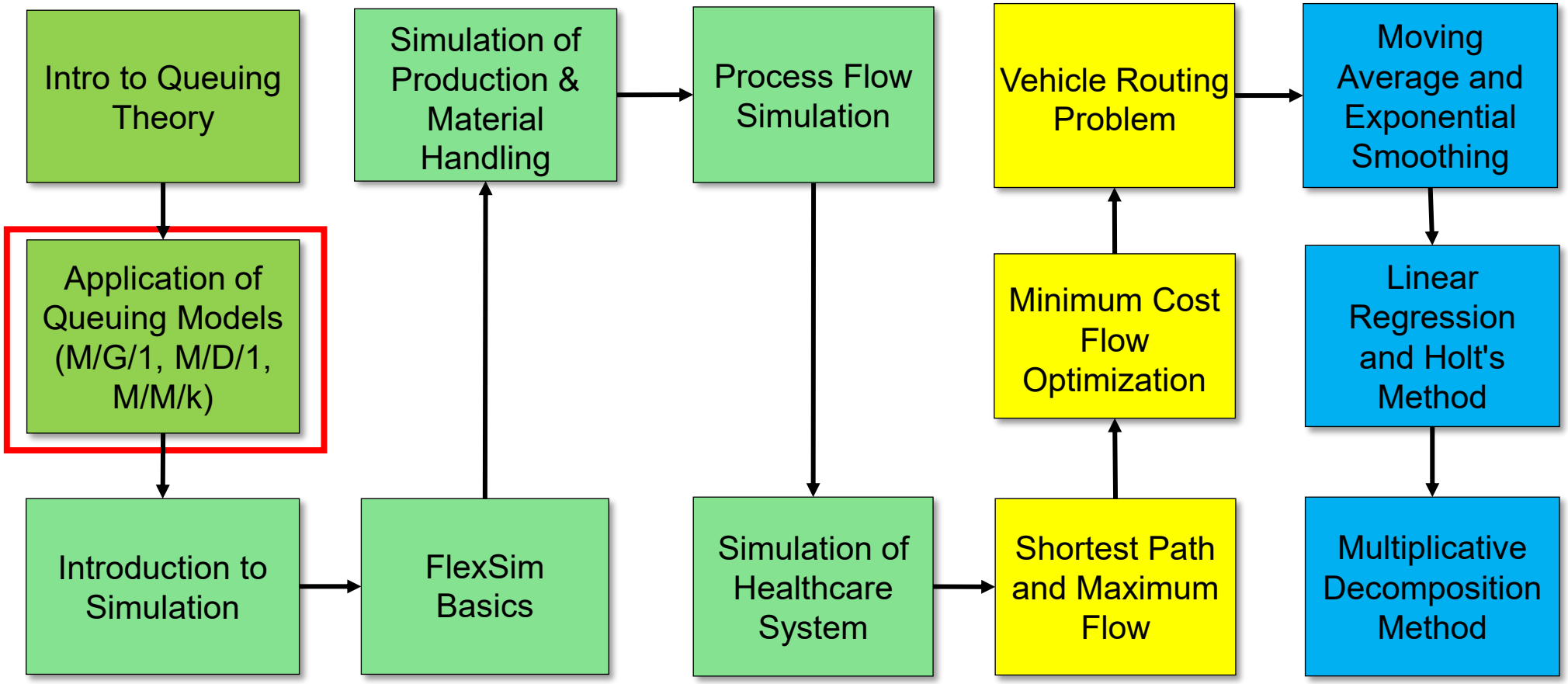
- The difference among M/M/1, M/G/1 and M/D/1 queuing models is the **variation of the service times**.
- **M/M/1** model represents a process with large service time variations (standard deviation = mean);
- **M/G/1** model represents a process with smaller service time variations that are represented by a specific standard deviation value;
- **M/D/1** represents a process with constant service time (standard deviation = 0)
- As the standard deviation of service times gets **smaller**, the performance of the queuing system becomes **better** → waiting time and queue length become shorter.
- If standard deviation of service times in a M/M/1 model is reduced to zero, the queuing system will become a M/D/1 model. Consequently, the waiting time and queue length will be halved.
- In other words, **Lq and Wq of M/D/1 model is always half of those of M/M/1 model.**

Learning Outcomes

At the end of the lesson, students should be able to:

- Apply Kendall's notation to classify a queuing system.
- Calculate performance measures of M/M/1, M/G/1 and M/D/1 queuing models.
- Explain how variability in service time affects the performance measures of a queuing system.

E211 Progression



SCORM Quiz

Scenario

At a convenience store, a cashier take on average 48 seconds to process a checkout and on average 63 customers arrive every hour.

Your task is to calculate the performance measures for this system under three different models: M/M/1, M/G/1, and M/D/1. You will apply the equations you've just learnt see how different assumptions about service time affect system performance.

Quiz

Click the **Quiz** button to edit this object

"At a convenience store, a cashier take on average 48 seconds to process a checkout and on average 63 customers arrive every hour."

Calculate the arrival rate of customers (in per minute). Enter only the numerical value in the box.